

→ Recall:

* List-set :: "a list $\Rightarrow$ 'a set"

$$\text{List-set } [] = \emptyset$$

$$\text{List-set } (x :: xs) = \text{insert } x (\text{List-set } xs)$$

* Tree-set

$$\text{Tree-set Leaf} = \emptyset$$

$$\text{Tree-set } (\text{Node } l \ x \ r) = \text{insert } x (\text{Tree-set } l \cup \text{Tree-set } r)$$

* inorder :: tree $\Rightarrow$ 'a list

$$\text{inorder Leaf} = []$$

$$\text{inorder } (\text{Node } l \ x \ r) = (\text{inorder } l) @ [x] @ (\text{inorder } r)$$

→ Prove $\text{List-set } (\text{inorder } t) = \text{tree-set } t$

proof by induction on t

base case . . . $t = \text{Leaf}$

assumptions: —

goal: $\text{List-set } (\text{inorder Leaf}) = \text{tree-set Leaf}$

$$1. \text{List-set } (\text{inorder Leaf}) = \text{List-set } [] \text{ by subst inorder.simps (1)}$$

$$2. \dots = \emptyset \text{ by subst List-set.simps (1)}$$

$$3. \dots = \text{tree-set Leaf} \text{ by tree-set.simps (1)} \checkmark$$

Step case

assumptions: I.H.₁ $\text{List-set } (\text{inorder } t_1) = \text{tree-set } t_1$

I.H.₂ $\text{List-set } (\text{inorder } t_2) = \text{tree-set } t_2$

goal: $\text{List-set } (\text{inorder } (\text{Node } t_1 \ a \ t_2)) = \text{tree-set } (\text{Node } t_1 \ a \ t_2)$

$$1. \text{List-set } (\text{inorder } (\text{Node } t_1 \ a \ t_2)) = \text{List-set } (\text{inorder } t_1 @ [a] @ \text{inorder } t_2) \text{ by subst inorder.simps (2)}$$

$$2. \dots = \text{List-set } (\text{inorder } t_1) \cup \text{List-set } ([a] @ \text{inorder } t_2) \text{ by subst Lem } [x \mapsto \text{inorder } t_1, y \mapsto [a] @ \text{inorder } t_2]$$

$$3. \dots = \text{List-set } (\text{inorder } t_1) \cup \text{List-set } [a] \cup \text{List-set } (\text{inorder } t_2)$$

by subst Lem₁

Now we can use I.H.₁ & I.H.₂

$$4. \dots = \text{tree-set } t_1 \cup \text{List-set } [a] \cup \text{List-set } (\text{inorder } t_2) \text{ by subst I.H.}_1$$

$$5. \dots = \text{tree-set } t_1 \cup \text{insert } a (\text{List-set } []) \cup \text{List-set } (\text{inorder } t_2) \text{ by List-set.simps (2)}$$

$$6. \dots = \text{tree-set } t_1 \cup \text{insert } a \emptyset \cup \text{List-set } (\text{inorder } t_2) \text{ by subst List-set (1)}$$

$$7. \dots = \text{tree-set } t_1 \cup [a] \cup \text{tree-set } t_2 \text{ by subst I.H.}_2$$

$$8. \dots = [a] \cup (\text{tree-set } t_1 \cup \text{tree-set } t_2)$$

$$9. \dots = \text{tree-set } (\text{Node } t_1 \ a \ t_2) \checkmark$$